VERSION 4.0 / RELEASE CHANGE REPORT

Administration & operations

A new protected admin console, seat and section controls, user role management, cross-user booking oversight, and development email notifications.

RELEASE	TAG	STACK
Administration milestone	v4.0	Flask · SQLAlchemy · SQLite · Flask-Mail

What Version 4 adds

Admin dashboard

Four KPI totals, recent reservation activity, consistent navigation and quick actions.

Event and seating administration

Existing full event CRUD plus block/unblock controls and create, rename and delete section workflows.

User role management

Admin-only promotion/demotion of student, staff and admin roles; self-edit and final-admin safeguards.

Booking administration

Paginated, filterable view across users, events and statuses; administrator cancellation preserves history.

Email notifications

Flask-Mail confirmation and cancellation messages, with console preview or optional Ethereal test SMTP.

Availability and database integrity

Blocked seats shown in the public map/API, with SQLite triggers enforcing blocked/active-booking consistency.

Preserved from V3

Public event pages; login and registration; student/staff/admin roles; event CRUD; responsive seat picker and five-second polling; atomic checkout; double-booking protection; personal booking history and owner cancellation.

Scope: Development-focused admin and messaging capabilities. No payment processing, tickets, refunds or guaranteed email delivery.

ARCHITECTURE / SECURITY

Access and data safeguards

Admin pages require a signed-in account with role admin. State-changing actions use POST and Flask-WTF CSRF protection.

Operational permissions

Area	Visitors / students / staff	Administrators
Dashboard & metrics	No access	KPIs, recent activity
Event catalog	Public browsing	Create, edit, safe delete
Seating	View, reserve unblocked seats	Block, unblock, manage sections
Roles	Public registration only makes student	Promote/demote other users
Booking history	View/cancel own reservations	View/filter all and cancel active

SQLite compatibility upgrade

- Adds seats.is_blocked, a non-null Boolean defaulting to false, to existing Version 3 tables without deleting seat or booking rows.
- Keeps the V3 partial unique index: at most one reserved booking per seat.
- Adds three SQLite triggers: reject reserved bookings on blocked seats, reject reactivation into blocked seats, and reject blocking an already reserved seat.
- Section rename retains seat IDs and booking relationships; section/event deletion refuses any booking history.

Email lifecycle

Event	Notification behavior
Multi-seat confirmation	One message per successful checkout after all booking rows commit.
Owner or admin cancellation	One message per cancelled booking sent to the booking owner after commit.
Already cancelled / validation fails	No repeated or false message.
Email transport error	Logged without removing or rolling back committed reservations.

Development mode: Console backend prints email previews. Ethereal SMTP can be configured using environment variables. No SMTP password is embedded in source code.

DELIVERY / OPERATIONS

Setup, verification & release

How to run Version 4

- 1. Extract CSTBS-v4.zip and enter CSTBS-v4/.
- 2. Create and activate a Python 3.10+ virtual environment.
- 3. Install dependencies with python -m pip install -r requirements.txt.
- 4. Copy .env.example to .env, replace SECRET_KEY with a generated random value, and leave MAIL_BACKEND=console for local previews.
- 5. Run python -m flask --app run.py seed-events, then python run.py.
- 6. Register a trusted account; promote it locally using python -m flask --app run.py set-role you@campus.edu admin.
- 7. Sign back in and navigate to http://127.0.0.1:5000/admin/.

Upgrading a V3 database

Stop the older app, make a backup of the entire SQLite file, and copy instance/cstbs.db into the new project before first launch. V4 runs targeted compatibility upgrades for the block flag and triggers; existing users, events, bookings and seat IDs are preserved. These scripts are development upgrades, not a full migration framework.

Validation status

Completed in this environment	Not verified here
Python syntax; Jinja template compilation; standalone SQLite blocking-trigger behavior; archive and PDF inspection.	Full Flask HTTP/integration tests and live SMTP delivery require dependencies and a configured local runtime.

Local test command: python -m unittest discover -s tests -v. The archive includes previous tests and the new V4 admin suite.

GitHub release description

Tag: v4.0 | Title: CSTBS v4.0 - Administration

Adds an admin dashboard, protected event and seat administration, section management, user-role controls, booking oversight/cancellation, console/Ethereal booking emails, and concurrency-safe SQLite blocked-seat constraints. Preserves V3 reservations and previous features. Development release: no payments or ticket issuance.

```
git add .  git commit -m "feat: introduce CSTBS v4.0 administration"
git tag -a v4.0 -m "CSTBS v4.0 - Administration"  git push origin main  git push origin v4.0
```